

Noah Thomas

(905) 730-6237 | nethomas@uwaterloo.ca

[linkedin.com/in/noah-thomas-7a13892a9](https://www.linkedin.com/in/noah-thomas-7a13892a9) | noah155.github.io | github.com/noah155

EDUCATION

University of Waterloo

Waterloo, ON

Mechanical Engineering, Option in Biomechanics (3.7/4.0 GPA) - BASc (Honours)

Sept. 2024 – Jun. 2029

PROJECTS

NeuroGrip | *SolidWorks, C++, Arduino, Electromyography, Bambu Studio, 3MF, DFM* Feb. 2026 – Mar. 2026

- Designed and fabricated a 3D printed dual tendon driven robotic hand using SolidWorks and Bambu Studio
- Drove prosthetic control using myoelectric (EMG) sensors and an Arduino UNO R3
- Implemented real-time EMG signal processing and threshold-based grip classification in C++
- Incorporated PWM driver board and 5 MG90D servos allowing for independent finger actuation
- MVP successfully able to conform to various objects, zero misclassifications across 30 trials

StrideSync | *C++, PlatformIO, GitHub, Python, ESP32, Soldering, SolidWorks, 3MF* Apr. 2026 – Present

- Wearable running gait analysis driven by six IMU sensors across both legs (per leg: thigh, shank, foot) measuring several key running metrics
- Validated MPU-6050 I²C data acquisition with Mahony filter
- Implemented TCA9548A multiplexer for multi-IMU SDA/SCL bus management
- Version-controlled via GitHub + VS Code + PlatformIO
- Dashboard implemented using Python with Plotly, Pandas, NumPy, and Matplotlib to visualize data effectively
- Next focuses include physical circuit & housing fabrication; anticipated completion in Summer '26

EXPERIENCE

Engineering Co-op

Jan. 2026 – May 2026

Mondelez International

Hamilton, ON

- Led execution efforts in \$350,000 process improvement project through effective management of budget and schedule while maintaining communications with a cross-functional team, resulting in yearly cost savings of \$110,000
- Prepared CFD geometry for \$1,000,000+ oven upgrades project by utilizing Autodesk Inventor to model oven airflow path, providing critical information on maximum fan operating speed which prevents dust cloud formation
- Brought employee locker room project from ideation to implementation in two months through layout mapping with AutoCAD and communication with contractors, increasing employee capacity by 20%

Assistant QA Inspector

May 2025 – Aug. 2025

S&C Electric Canada Ltd.

Toronto, ON

- Identified manufacturing defects of high voltage electrical switches through evaluation of engineering drawings using GD&T, ensuring dimensional compliance to engineering specifications and safety-critical design standards
- Ensured the accuracy of measurement gauges by using metrology principles including calibration and maintenance, resulting in the company wide retention of ISO 9001 certification
- Quarantined non-conforming parts through production inspections, eliminating 20 hours of downtime bimonthly

VOLUNTEERING

Dominican Republic Education and Medical Support (DREAMS)

Jan. 2025 – Apr. 2025

ADEJSO

CAN & DOM

- Conducted various fundraising efforts, resulting in +\$10,000 raised for underprivileged families throughout the province of San Jose de Ocoa within the Dominican Republic
- Collaborated with peers in the construction of a house while staying within La Yuca for a week
- Distributed essential first aid supplies and over-the-counter medications (including pain relievers, antihistamines, and anti-nausea treatments) to support the local community

TECHNICAL SKILLS

CAD/MFG: SolidWorks, Inventor, AutoCAD, Bambu Studio

Languages: C, C++, Python, MATLAB + Simulink, Simscape Multibody

Toolkits: VS Code, PlatformIO, Git/GitHub, Docker (debian)

Hardware & Related Skills: Arduino UNO R3, ESP32, I²C protocols, Soldering